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#define TRIG_PIN 14      // Trigger pin of ultrasonic sensor
#define ECHO_PIN 13      // Echo pin of ultrasonic sensor
#define RELAY_PIN 12     // Relay pin
int LedPin =2;
long duration;
int distance;
int thresholdDistance = 10; // Distance threshold in cm for triggering the
relay

void setup() {
  pinMode(TRIG_PIN, OUTPUT); // Set the trigger pin as output
  pinMode(ECHO_PIN, INPUT);  // Set the echo pin as input
  pinMode(RELAY_PIN, OUTPUT); // Set the relay pin as output
  digitalWrite(RELAY_PIN, LOW); // Keep the relay off initially
  pinMode(LedPin, OUTPUT);
  Serial.begin(115200); // Initialize serial communication for debugging
  Serial.println("Train Monitor Initialized");
}

void loop() {
  // Ultrasonic sensor logic
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  // Read the duration of the pulse from the echo pin
  duration = pulseIn(ECHO_PIN, HIGH);

  // Calculate the distance in centimeters
  distance = duration * 0.034 / 2;

  // Display the distance on the Serial Monitor
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  if (distance <= thresholdDistance && distance > 0) {
    // If an object is detected within the threshold distance, turn on the relay
    digitalWrite(RELAY_PIN, HIGH);
    Serial.println("Train on Track - Relay ON");
    digitalWrite(LedPin, HIGH); // turn the LED on (HIGH is the voltage level)
    delay(500);
    delay(2500);
  } else {
    // If no object is detected, turn off the relay
    digitalWrite(RELAY_PIN, LOW);
    Serial.println("No Train - Relay OFF");
    digitalWrite(LedPin, HIGH); // turn the LED on (HIGH is the voltage level)
    delay(500); // wait for a second
    digitalWrite(LedPin, LOW); // turn the LED off by making the voltage LOW
    delay(500);
  }

  delay(500); // Wait for half a second before measuring again
}

```